Summary: Solar Flares and Soft/Hard X-rays

Solar flares are explosive releases of magnetic energy from the Sun caused by magnetic reconnection. They emit soft and hard X-rays that help scientists understand flare evolution and improve space weather forecasting.

Diagram 1: Simplified Solar Flare

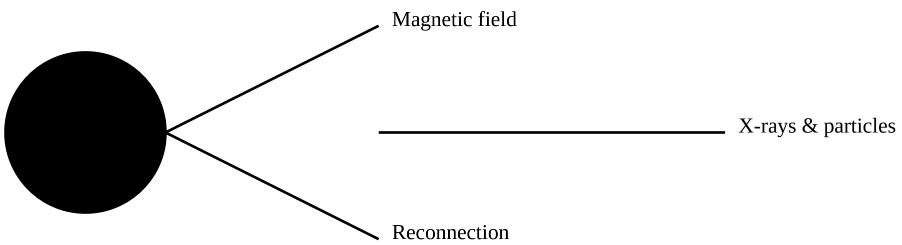
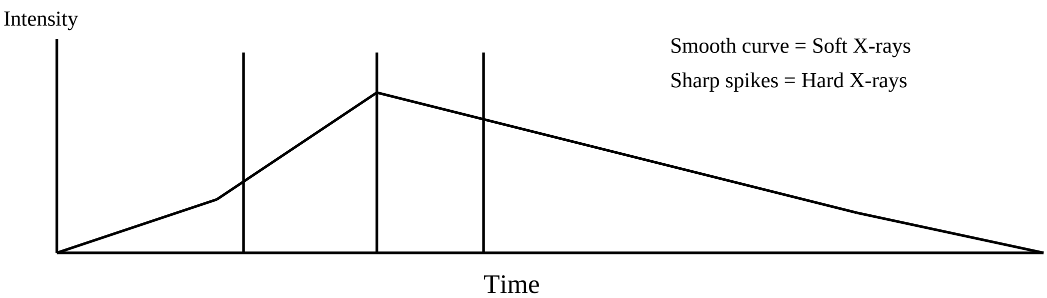


Diagram 2: Soft vs Hard X-ray Emission



Feature	Soft X-rays	Hard X-rays
Energy	0.1-12 keV	>10-120 keV
Origin	Hot flare loops	Footpoints / energetic electrons
Graph	Smooth rise & decay	Short impulsive spikes

Key Point: According to the Neupert Effect, hard X-ray bursts occur during the rapid rise of soft X-rays, indicating energetic electrons heat the plasma that later emits soft X-rays.